
Tackling Subpopulation Attacks: Evaluating Mitigation Strategies for Subpopulation Attacks in Machine Learning

Abstract

The integrity and dependability of machine learning models are severely threatened by data poisoning attacks, especially in situations where adversaries alter training data to cause misclassification. Among those poisoning attacks, subpopulation attacks are one type of attack that targets particular subsets of data in order to undermine model performance within those subpopulations. In order to restore data integrity and model robustness, ten mitigation techniques are proposed in this research, which also looks into subpopulation attacks. Initially, a purposeful poisoning of a dataset is applied to mimic adversarial real-world situations. Next, ten mitigation strategies are assessed for effectiveness using both comparative and practical methods. In the practical method, poisoned data is cleaned using machine learning models, and each method's effectiveness is evaluated separately. In the meantime, the comparative method entails comparing these techniques' performance to a standard. Moreover, the mitigation process makes use of machine learning models, which use a variety of algorithms to clean up contaminated data and improve model performance. These models' performance is carefully assessed, taking into account various metrics including F1-score, recall, accuracy, and precision. This work provides a thorough analysis of subpopulation attacks and mitigation strategies, which advances data security in machine learning environments. The results offer significant perspectives on the efficacy of various mitigation tactics, enabling well-informed decision-making when safeguarding against attacks utilizing data poisoning.